

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Industry 4.0

Section Id :	640653138800
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	21
Number of Questions to be attempted :	21
Section Marks :	45
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653386979
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532125106 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED

EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536925425. ✓ Yes

6406536925426. ✗ No

Sub-Section Number : 2

Sub-Section Id : 640653386980

Question Shuffling Allowed : No

Question Id : 6406532125107 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (2 to 3)

Question Label : Comprehension

The owner of a newsstand wants to determine the number of newspapers of *USA Now* to be stocked at the start of each day. The owner pays 30 cents for a copy and sells it for 75 cents. The sale of the newspaper typically occurs between 7:00 and 8:00 am. Newspaper left at the end of the day are recycled for an income of 5 cents a copy. How many copies should the owner stock every morning, assuming that the demand for the day can be described as:

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 2 Question Id : 6406532125108 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

How many copies should the owner stock each morning, assuming that the daily demand follows a normal distribution with a mean of 300 copies and a standard deviation of 20 copies?

[Standard_Norm_inv(critical fractile)=0.643]

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

307 to 313

Question Number : 3 Question Id : 6406532125109 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

How many copies should the owner stock each morning, assuming that the daily demand follows a discrete probability distribution, $f(D)$, defined as follows?

D	200	220	300	320	340
$f(D)$	0.1	0.2	0.4	0.2	0.1

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

300

Question Id : 6406532125110 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (4 to 5)

Question Label : Comprehension

Suppose you need to accept reservations for a flight with a total of 100 seats available. There are two classes of passengers: economy and business. An economy seat generates ₹2,500 in revenue, while a business seat generates ₹7,500. Based on past experience, the demand for business travellers follows a uniform distribution between 10 and 40. Answer the given subquestions.

Sub questions

Question Number : 4 Question Id : 6406532125111 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

What is the Booking limit?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 5 Question Id : 6406532125112 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

What is the protection level?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

70

Question Id : 6406532125116 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (6 to 7)

Question Label : Comprehension

A store sells a seasonal product with demand following a normal distribution, having a mean of 500 units and a standard deviation of 75 units. Each unit is purchased for ₹200 and sold for ₹350. Any leftover stock is cleared at a discounted price of ₹100 per unit.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 6 Question Id : 6406532125117 Question Type : SA Keyboard Layout :

Inscript

Correct Marks : 2

Question Label : Short Answer Question

Determine the optimal order quantity. (Z -value = 0.25)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

518 to 520

Question Number : 7 Question Id : 6406532125118 Question Type : SA Keyboard Layout :

Inscript

Correct Marks : 2

Question Label : Short Answer Question

If the store orders 520 units, what is the probability of a stockout? (probability corresponding to the Z -value = 0.06064) (enter your answer without the percentage symbol and rounded off to two decimals. E.g. if your answer is 39.223%, enter your answer as 39.22)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

38 to 40

Sub-Section Number : 3

Sub-Section Id : 640653386981

Question Shuffling Allowed : No

Question Id : 6406532125113 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (8 to 9)

Question Label : Comprehension

The Egg Dropping Problem is a Dynamic Programming (DP) problem that helps determine the minimum number of attempts needed to find the highest floor from which an egg can be dropped without breaking.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 8 Question Id : 6406532125114 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the state variable here?

Options :

6406536925431. ✓ Number of eggs on hand

6406536925432. ✗ Number of floors

6406536925433. ✗ Number of eggs that got broken

Question Number : 9 Question Id : 6406532125115 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How a stage is defines

Options :

6406536925434. ✗ Number of eggs on hand

6406536925435. ✓ Number of floors

6406536925436. ✗ Number of eggs that got broken

Sub-Section Number : 4

Sub-Section Id : 640653386982

Question Shuffling Allowed : No

Question Id : 6406532125119 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (10 to 12)

Question Label : Comprehension

Rahul owns a tea distribution business in Assam. He sources premium tea leaves from local tea gardens and supplies them to various retailers. To manage his inventory efficiently, he wants to determine the optimal order quantity for his stock.

- i) The annual demand for tea leaves is 24,000 kg.
- ii) The cost of placing an order (ordering cost) is ₹500 per order.
- iii) The carrying cost per kg of tea per year is ₹2.

Rahul wants to minimize his total inventory cost by ordering the optimal quantity each time. Answer the given three subquestions

Sub questions

Question Number : 10 Question Id : 6406532125120 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

Calculate the Economic Order Quantity (EOQ) for Rahul's business.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

5770 to 5777

Question Number : 11 Question Id : 6406532125121 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

How many orders should Rahul place per year?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

6 to 8

Question Number : 12 Question Id : 6406532125122 Question Type : SA Keyboard Layout :

Inscript

Correct Marks : 2

Question Label : Short Answer Question

What is the total cost associated with holding inventory at EOQ?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3455 to 3475

Sub-Section Number : 5

Sub-Section Id : 640653386983

Question Shuffling Allowed : Yes

Question Number : 13 Question Id : 6406532125123 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Find the Euclidean distance between the points A(3, 4) and B(7, 1) in a 2D plane.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number : 6

Sub-Section Id : 640653386984

Question Shuffling Allowed : Yes

Question Number : 14 Question Id : 6406532125124 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

Rana Bakery, a company, uses a layout consisting of 9 workstations. Each day, 200 sandwiches are produced using this layout. A typical workday consists of a single 10-hour shift. The sandwich-making process comprises 9 activities, as detailed in the table below.

Activity ID	Activity Description	Activity Time (in seconds)	Preceding activities
A	Wheat bag emptying	20	
B	Vegetable cleaning	10	
C	Bread baking	60	A
D	Vegetable cutting	40	B
E	Bread Slicing	20	C
F	Butter applying	5	E
G	Vegetable staffing	5	E, F
H	Sandwich baking	15	G
I	Sandwich packing	5	H

What is the current efficiency of the line where each activity is performed at a separate workstation? (Enter only the numerical value up to two decimal places without the '%' symbol. For example, if the answer is 99.756%, enter only '99.76').

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Range
Text Areas : PlainText
Possible Answers :

11.10 to 11.12

Sub-Section Number : 7
Sub-Section Id : 640653386985
Question Shuffling Allowed : Yes

Question Number : 15 **Question Id :** 6406532125125 **Question Type :** MSQ
Correct Marks : 2 **Max. Selectable Options :** 0

Question Label : Multiple Select Question
 Which of the following supply chain costs is/are affected by the distribution network structure?

- Options :**
- 6406536925444. ✓ Inventory
 - 6406536925445. ✓ Transportation
 - 6406536925446. ✓ Facilities and handling
 - 6406536925447. ✓ Information
 - 6406536925448. ✗ Price

Question Number : 16 **Question Id :** 6406532125126 **Question Type :** MSQ
Correct Marks : 2 **Max. Selectable Options :** 0

Question Label : Multiple Select Question
 Select the correct statements from the list below:
Options :

6406536925449. ✖ Optimal solution in a facility location problem using Euclidean method will result in a region

6406536925450. ✔ Optimal solution in a facility location problem using Euclidean method rarely matches the center of gravity

6406536925451. ✔ Optimal solution in a facility location problem using Euclidean method will result in a point

6406536925452. ✖ Optimal solution in a facility location problem using Euclidean method matches the centre of gravity in not of the cases

Question Number : 17 Question Id : 6406532125127 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

What are the two primary tasks of time series forecasting?

Options :

6406536925453. ✔ Identifying pattern

6406536925454. ✖ Identify variables of interest

6406536925455. ✔ Discovering relationships amongst variables of interest

Question Number : 18 Question Id : 6406532125128 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Identify the examples for Outsourcing from the list below:

Options :

6406536925456. ✖ Micron Technology, a US based semiconductor giant, setting up a Supply Chain Analytics team, in India.

6406536925457. ✖ Mastercard has set up a Data Science team in Gurgaon to develop algorithms for the company.

6406536925458. ✔ IIT Madras' BS degree program has assigned TCS to develop and manage the exam portal and its evaluation.

6406536925459. ✔ Foxconn assembles iPhones for Apple

Sub-Section Number : 8

Sub-Section Id : 640653386986

Question Shuffling Allowed : Yes

Question Number : 19 Question Id : 6406532125129 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What does the decision variable $X_{i,j}$ denote in a Vehicle Routing Problem (VRP)?

Options :

6406536925460. ✔ If the vehicle travels to node j from node i (binary)

6406536925461. ✖ Number of vehicles travelled between node i and node j (integer)
6406536925462. ✖ Weight on the arc (i,j) e.g. the distance (integer)

Question Number : 20 Question Id : 6406532125130 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following is not an example of digitization?

Options :

6406536925463. ✖ Creating the e-copy of a book using Google-Lens
6406536925464. ✔ Generating an automated daily sales report across the retail outlets in a state.
6406536925465. ✖ Converting a physical 2D sheet to CAD drawing
6406536925466. ✖ Converting physical invoice to e-copy

Question Number : 21 Question Id : 6406532125131 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Match Column A with Column B for landmark outcomes of the industrial era.

Col A (Industrial Era)	Col B (Major Outcome)
1.0	Digitization
2.0	CNC
3.0	Steam Engine
4.0	Mass production

Options :

6406536925467. ✖ 1.0:CNC, 2.0:Mass production, 3.0:Digitization, 4.0:Steam Engine
6406536925468. ✖ 1.0:Steam Engine, 2.0:Mass production, 3.0:Digitization, 4.0:CNC
6406536925469. ✖ 1.0:Mass production, 2.0:Steam Engine, 3.0:Digitization, 4.0:CNC
6406536925470. ✔ 1.0:Steam Engine, 2.0:Mass production, 3.0:CNC, 4.0:Digitization

Question Number : 22 Question Id : 6406532125132 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following is an advantage of a Process Layout?

Options :

6406536925471. ✖ Low material handling
6406536925472. ✖ Total production time per unit is small
6406536925473. ✖ Low training cost for the operators
6406536925474. ✔ It can accommodate high variety of products

Question Number : 23 Question Id : 6406532125134 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following functions does Logistics Management not cover?

Options :

6406536925479. ✖ Supply Management

6406536925480. ✖ Materials Management

6406536925481. ✔ Customer Management

6406536925482. ✖ Distribution Management

Question Number : 24 Question Id : 6406532125135 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following is not a form of internet?

Options :

6406536925483. ✖ Industrial

6406536925484. ✖ Commercial

6406536925485. ✔ Philanthropic

6406536925486. ✖ Consumer

Sub-Section Number :

9

Sub-Section Id :

640653386987

Question Shuffling Allowed :

Yes

Question Number : 25 Question Id : 6406532125133 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the name of the forecasting method represented by the following set of equations? The notations have their usual meanings.

$$F_{t+1} = a_t + b_t$$

Where

$$a_t = \alpha D_t + (1 - \alpha)(a_{t-1} - b_{t-1});$$

$$b_t = \beta(a_t - a_{t-1}) + (1 - \beta)b_{t-1}$$

Options :

6406536925475. ✖ Exponential smoothing

6406536925476. ✖ Winter's model for seasonality

6406536925477. ✔ Holt's method for trend

6406536925478. ✖ None of these

Question Number : 26 Question Id : 6406532125136 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

You visit the car service center to get your car washed. In the service center, washing done using automated systems, e.g. the car is put on a moving belt, water is sprayed, drying is also done. Which of the following terms represents this scenario?

Options :

6406536925487. ✖ Digitization

6406536925488. ✔ Automation

6406536925489. ✖ Digital transformation

Question Number : 27 Question Id : 6406532125137 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is not an example of digitization ?

Options :

6406536925490. ✖ Scanning a document to get the PDF document

6406536925491. ✖ Converting a physical 2D sheet to CAD drawing

6406536925492. ✖ Converting physical invoice to e-copy

6406536925493. ✔ Creating an automated report from the existing cost data